AN INVERSE PROBLEM FOR CONVOLUTION MEASURES WITH APPLICATIONS

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Abstract.

1. Introduction

A classical theorem of Jörgens ($d = 2$), Calabi ($d \leq 5$) and Pogorelov ($d \geq 2$) states that any classical convex solution of $\det D^2 u = 1$ in $\mathbb{R}^d$ must be a quadratic polynomial. As stated in the second paragraph of [1], Caffarelli extended the result from classical solutions to *viscosity solutions*. We recall the definitions.

Definition 1 ([6])**.** *Let $u \in C^0(\Omega)$ be a convex function and let $0 \leq f \in C^0(\Omega)$. The function u is a viscosity subsolution (resp. supersolution) of the equation $\det D^2 u = f$ in Ω if whenever convex $\phi \in C^2(\Omega)$ and $x_0 \in \Omega$ are such that $(u - \phi)(x) \leq (u - \phi)(x_0)$ (resp. $\geq$) for all x in a neighborhood of x_0, then necessarily*

$$\det D^2\phi(x_0) \geq f(x_0) \qquad (\textit{resp.} \ \ \leq).$$

As in [6, Remark 1.3.3], no generality is lost in assuming that $\phi(x_0) = u(x_0)$, and one may further restrict the test functions to strictly convex quadratic polynomials.

Let $\psi : \mathbb{R}^d \to \mathbb{R}$ be convex, and let σ denote projection measure on the graph

$$\{(x, \psi(x)) : x \in \mathbb{R}^d\} \subset \mathbb{R}^{d+1}.$$

Thus

$$\langle \sigma, \varphi \rangle = \int_{\mathbb{R}^d} \varphi(x, \psi(x)) \, \mathrm{d}x.$$

The centered convolution quantity under scrutiny is

$$F_\sigma(\xi, \tau) := (\sigma * \sigma)(\xi, 2\psi(\xi/2) + \tau), \qquad \tau > 0.$$

Via the usual change of variables from [10, Prop. 2.1(c)] and [11, Prop. 2.1(c)], this can be written as

$$F_\sigma(\xi, \tau) = \int_{\mathbb{R}^d} \boldsymbol{\delta}\big(\tau - \Psi(\xi, y)\big) \, \mathrm{d}y, \qquad \Psi(\xi, y) := \psi(\xi/2 + y) + \psi(\xi/2 - y) - 2\psi(\xi/2). \tag{1}$$

Convexity gives $\Psi \geq 0$. If ψ is strictly convex, then $\Psi(\xi, y) > 0$ whenever $y \neq 0$.

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For a quadratic polynomial

$$\psi(x) = a + b \cdot x + \frac{1}{2} x^T H x, \qquad H = H^T > 0,$$

one has

$$\Psi(\xi, y) = y^T H y$$

for every ξ, y. Consequently

$$F_\sigma(\xi, \tau) = \frac{1}{\sqrt{\det H}} \int_{\mathbb{R}^d} \boldsymbol{\delta}\big(\tau - |w|^2\big) \, \mathrm{d}w = \frac{d\,\omega_d}{2\sqrt{\det H}} \tau^{d/2-1},$$

where $\omega_d = |\{x \in \mathbb{R}^d : |x| < 1\}|$. Thus the constant profile from the two-dimensional paraboloid is a special feature of dimension two; in higher dimensions a quadratic graph has the power-law profile $\tau^{d/2-1}$.

The point of the argument below is that the integrated form of the convolution identity is stronger than the almost everywhere Hessian information discussed in the transcript. It tests ψ against upper and lower quadratic supports and therefore gives a viscosity Monge–Ampère equation directly, once the centered sections are known to shrink to their center. This last point is automatic under strict convexity, but we isolate it below because it is the only place where strict convexity is used.

2. An inverse result

Definition 2. *Let $c_0 > 0$. We say that the centered convolution is constant on its normalized support with constant c_0 if, for every $\xi \in \mathbb{R}^d$ and every $t > 0$,*

$$|\{y \in \mathbb{R}^d : \Psi(\xi, y) < t\}| = c_0 t. \tag{2}$$

Equivalently, after differentiating in t in the sense of distributions for each fixed ξ,

$$F_\sigma(\xi, \tau) = c_0, \qquad \tau > 0. \tag{3}$$

Definition 3. *We say that ψ has shrinking centered sections if, for every $x \in \mathbb{R}^d$ and every $r > 0$,*

$$\min_{|y|=r} \big(\psi(x+y) + \psi(x-y) - 2\psi(x)\big) > 0.$$

This is automatic if ψ is strictly convex.

Theorem 4 (Constant normalized support)**.** *Let $d \geq 1$, let $\psi : \mathbb{R}^d \to \mathbb{R}$ be convex, and let σ be projection measure on its graph. Assume that the centered convolution is constant on its normalized support with constant $c_0 > 0$. Assume moreover that:*

1. *ψ has a nondegenerate Alexandrov second-order expansion at least one point;*
2. *ψ has shrinking centered sections.*

Then $d = 2$, and

$$\psi(x) = a + b \cdot x + \frac{1}{2} x^T H x$$

for some $a \in \mathbb{R}$, $b \in \mathbb{R}^d$, and some positive definite symmetric matrix H, with

$$\det H = \left(\frac{\pi}{c_0}\right)^2. \tag{4}$$

After subtracting the affine part and making the linear change of variables which sends H to $2I$, this is exactly the model $\psi(x) = |x|^2$.

Conversely, every such positive definite quadratic polynomial in dimension two satisfies (3) with $c_0 = \pi/\sqrt{\det H}$. In particular, $\psi(x) = |x|^2$ gives $c_0 = \pi/2$.

Remark 5. *The exact conclusion $\psi(x) = |x|^2$ cannot follow from constancy alone unless one fixes the affine and linear normalizations. Indeed every positive definite quadratic polynomial in dimension two has constant centered convolution; the constant only sees $\det H$, not the eccentricity of H.*

Lemma 6 (Local section asymptotics)**.** *Let $\phi \in C^2$ be convex in a neighborhood of $x_0 \in \mathbb{R}^d$, and set*

$$Q(y) := \phi(x_0 + y) + \phi(x_0 - y) - 2\phi(x_0), \qquad A := D^2\phi(x_0).$$

If A is positive definite, then, for every $r > 0$,

$$\lim_{t\to 0^+} t^{-d/2}|\{y \in B(0,r) : Q(y) < t\}| = \frac{\omega_d}{\sqrt{\det A}}. \tag{5}$$

If A is singular, then, for every $r > 0$,

$$\lim_{t\to 0^+} t^{-d/2}|\{y \in B(0,r) : Q(y) < t\}| = +\infty \tag{6}$$

.

Proof. The positive definite case follows from

$$Q(y) = y^T Ay + o(|y|^2)$$

and the linear change of variables $w = A^{1/2}y$; for $t > 0$ sufficiently small, the relevant section is contained in $B(0,r)$.

If A is singular, fix $r > 0$ and split $\mathbb{R}^d = E \oplus K$, where $K = \ker A$ has dimension $k \geq 1$ and $E = K^\perp$. Since $Q(y) = y^T Ay + o(|y|^2)$, for every $\varepsilon > 0$ there is a radius $0 < \rho < r$ such that, for $y = e + z \in E \oplus K$ with $|y| < \rho$,

$$Q(y) \leq 2\, e^T Ae + \varepsilon|z|^2.$$

For $t > 0$ sufficiently small, the set

$$\{e \in E : 2e^T Ae < t/2\} \times \{z \in K : \varepsilon|z|^2 < t/2\}$$

is contained in $\{Q < t\} \cap B(0,r)$ and has measure $C_A\varepsilon^{-k/2}t^{d/2}$. Since ε is arbitrary, the normalized local section volume tends to $+\infty$. □

3. Proof of the characterization

Proof of Theorem 4. We first pin down the dimension. Let u_0 be a point at which ψ has a nondegenerate Alexandrov second-order expansion,

$$\psi(u_0+y)=\psi(u_0)+p\cdot y+\frac{1}{2}y^T H_0 y+o(|y|^2), \qquad H_0=H_0^T>0.$$

Then

$$\Psi(2u_0,y)=y^T H_0 y+o(|y|^2),$$

and therefore

$$|\{y\in\mathbb{R}^d:\Psi(2u_0,y)<t\}|\sim\frac{\omega_d}{\sqrt{\det H_0}}t^{d/2} \qquad (t\to 0^+).$$

Comparison with (2) forces $d=2$ and

$$\det H_0=\left(\frac{\pi}{c_0}\right)^2.$$

It remains to prove that ψ is quadratic. Set

$$\lambda:=\left(\frac{\pi}{c_0}\right)^2.$$

We prove that ψ is a viscosity solution of

$$\det D^2\psi=\lambda \qquad \text{on } \mathbb{R}^2. \tag{7}$$

Let $\phi\in C^2$ be convex near x_0, with

$$\phi(x_0)=\psi(x_0), \qquad \phi\geq\psi$$

near x_0. Choose $r>0$ so small that the comparison holds whenever $|y|<r$. For such y,

$$\Psi(2x_0,y)\leq\phi(x_0+y)+\phi(x_0-y)-2\phi(x_0)=:Q(y).$$

Hence, for every $t>0$,

$$\{y\in B(0,r):Q(y)<t\}\subset\{y\in\mathbb{R}^2:\Psi(2x_0,y)<t\}.$$

Thus

$$|\{y\in B(0,r):Q(y)<t\}|\leq c_0 t.$$

In view of Lemma 6, the matrix $A=D^2\phi(x_0)$ cannot be singular. Therefore (5) gives

$$\frac{\pi}{\sqrt{\det A}}\leq c_0,$$

or equivalently

$$\det D^2\phi(x_0)\geq\lambda.$$

This is the viscosity subsolution inequality.

Now let $\phi\in C^2$ be convex near x_0, with

$$\phi(x_0)=\psi(x_0), \qquad \phi\leq\psi$$

near x_0. Choose $r > 0$ so small that the comparison holds whenever $|y| < r$. By the shrinking-section assumption,

$$m_r := \min_{|y|=r} \Psi(2x_0, y) > 0,$$

and $\{\Psi(2x_0, \cdot) < t\} \subset B(0, r)$ whenever $0 < t < m_r$. Indeed, any y with $|y| > r$ and $\Psi(2x_0, y) < m_r$ would give, by convexity along the ray from 0 to y,

$$\Psi(2x_0, ry/|y|) \leq \frac{r}{|y|}\Psi(2x_0, y) < m_r,$$

which is impossible. Hence

$$\{y \in \mathbb{R}^2 : \Psi(2x_0, y) < t\} \subset \{y \in B(0, r) : Q(y) < t\}$$

whenever $0 < t < m_r$. If $A = D^2\phi(x_0)$ is singular, then $\det A = 0 \leq \lambda$. If A is positive definite, then

$$c_0 t \leq |\{y \in B(0, r) : Q(y) < t\}|$$

for $0 < t < m_r$, and (5) implies

$$c_0 \leq \frac{\pi}{\sqrt{\det A}},$$

or equivalently

$$\det D^2\phi(x_0) \leq \lambda.$$

This is the viscosity supersolution inequality.

Thus ψ is a convex viscosity solution of (7). By the Jörgens theorem in the viscosity form of Caffarelli–Li [1, Theorem 1.1], after the harmless normalization of the right-hand side of (7) to 1, ψ is a quadratic polynomial. This proves the direct implication and the determinant identity (4).

For the converse, let

$$\psi(x) = a + b \cdot x + \frac{1}{2}x^T H x, \qquad H = H^T > 0.$$

Then $\Psi(\xi, y) = y^T H y$, independently of ξ. Consequently

$$F_\sigma(\xi, \tau) = \int_{\mathbb{R}^2} \boldsymbol{\delta}\big(\tau - y^T H y\big)\, \mathrm{d}y = \frac{\pi}{\sqrt{\det H}},$$

which is (3) with $c_0 = \pi/\sqrt{\det H}$. □

4. The degenerate alternative

We now spell out what should replace the nondegeneracy hypothesis in higher dimensions. The dimension reduction in the proof of Theorem 4 used only one positive definite Alexandrov second-order expansion. In dimension $d \geq 3$ such a point is impossible under the linear section law (2): if

$$\psi(x_0 + y) = \psi(x_0) + p \cdot y + \frac{1}{2}y^T H_0 y + o(|y|^2), \qquad H_0 = H_0^T > 0,$$

then

$$|\{y \in \mathbb{R}^d : \Psi(2x_0, y) < t\}| \sim \frac{\omega_d}{\sqrt{\det H_0}} t^{d/2}, \qquad t \to 0^+,$$

which is incompatible with $c_0 t$ when $d \neq 2$.

In the rest of this section we also impose the lower normalization

$$\psi \geq 0.$$

This is the normalization relevant to the final geometric obstruction.

Thus, in dimensions $d \geq 3$, the only possible convex case is the completely degenerate Alexandrov alternative:

$$\det D^2\psi(x) = 0 \quad \text{at every Alexandrov point } x.$$

What is needed here is a developability statement for convex graphs. In the smooth complete case, the classical mechanism is the Hartman–Nirenberg theorem: a complete C^2 hypersurface of constant zero Gauss–Kronecker curvature in Euclidean space is cylindrical, i.e. it is ruled by affine $(d-1)$-planes [7]. For nonsmooth convex hypersurfaces, the natural language is Aleksandrov–Pogorelov bounded extrinsic curvature and the associated spherical-image curvature measure; see Pogorelov's monograph [12]. Ushakov gives a modern smooth account of developable surfaces in arbitrary dimension and codimension [14]. The expected form needed here is:

> If a finite convex function $\psi : \mathbb{R}^d \to \mathbb{R}$ has vanishing extrinsic curvature measure, and its graph is not an affine hyperplane, then through every point $(x, \psi(x))$ of the graph there is a nontrivial affine segment; under the global hypotheses relevant here, this segment should promote to a complete affine line in the graph.

The curvature-measure formulation is important. The a.e. Alexandrov degeneracy condition alone does not exclude singular curvature mass; it is therefore too weak to force rulings through every point. Equivalently, for every $x \in \mathbb{R}^d$ one should find a nonzero direction $v_x \in \mathbb{R}^d$ such that the one-dimensional slice

$$s \mapsto \psi(x + sv_x)$$

is affine on all of $\mathbb{R}$. In that case

$$\Psi(2x, sv_x) = 0 \qquad \text{for every } s \in \mathbb{R}.$$

This by itself is not yet a contradiction, since a line has zero d-dimensional measure. The required final step is a quantitative "thickening of rulings" argument: the presence of a complete flat direction through each point of the graph should force the centered sections

$$\{y : \Psi(2x, y) < t\}$$

to contain enough transverse thickness around that line to have either infinite volume or a small-t power law different from t. In the ideal cylindrical case, for example, the section would contain a product of the flat line with a transverse $(d-1)$-dimensional section, so the volume would be infinite. In less rigid ruled situations, one should still aim to prove a lower bound contradicting the exact linear identity (2).

This would finish the desired higher-dimensional contradiction. More precisely, the remaining task is to supply or cite the appropriate developability theorem in the low regularity class at hand. In the C^2 case this is the familiar flat-direction argument: $\det D^2\psi \equiv 0$ and convexity force a ruling by affine directions. For merely convex functions, one has to rule out singular Monge–Ampère mass, justify that Alexandrov degeneracy gives actual affine rulings of the graph rather than only a.e. flat directions, and then prove the quantitative thickening estimate for the centered sections.

4.1. **The convolution identity forces the zero-curvature hypothesis.** There is a slightly better way to phrase the remaining low-regularity assumption. We should not assume that the curvature measure vanishes. We should assume only that ψ belongs to a class in which the appropriate curvature measure is defined, and then derive its vanishing from (2).

Here is the argument. Let $d \geq 3$, fix $x_0 \in \mathbb{R}^d$, and let $\phi \in C^2$ be convex near x_0, with

$$\phi(x_0) = \psi(x_0), \qquad \phi \leq \psi$$

near x_0. As in the viscosity supersolution part of the proof of Theorem 4, assume that the centered sections shrink to their center, so that for $t > 0$ sufficiently small the set $\{\Psi(2x_0, \cdot) < t\}$ is contained in the neighborhood where the comparison with ϕ holds. Then

$$\{\Psi(2x_0, \cdot) < t\} \subset \{y : Q(y) < t\}, \qquad Q(y) := \phi(x_0 + y) + \phi(x_0 - y) - 2\phi(x_0).$$

If $A = D^2\phi(x_0)$ were positive definite, Lemma 6 would give

$$|\{y : Q(y) < t\}| = O(t^{d/2}), \qquad t \to 0^+.$$

This contradicts (2), because $d/2 > 1$ and $c_0 t \neq O(t^{d/2})$. Hence every lower quadratic test has singular Hessian:

$$\det D^2\phi(x_0) = 0.$$

In other words, ψ is a convex viscosity supersolution of

$$\det D^2\psi = 0.$$

Since the Monge–Ampère measure of a convex function is nonnegative, this is equivalent, in the Aleksandrov/viscosity correspondence, to vanishing Monge–Ampère measure:

$$M_\psi = 0.$$

Thus, if we work in a class where the graph curvature measure is the Aleksandrov–Pogorelov spherical-image measure, or in a class where $\operatorname{Det} D^2\psi$ is defined weakly, the vanishing condition is not an extra hypothesis. It is forced by the convolution identity, together with the shrinking-section localization. For graphs of convex functions, vanishing of M_ψ is the same zero-curvature condition after applying the Gauss-map parametrization of the spherical image; the weight relating the two measures is positive on bounded subgradient ranges.

4.2. **Possible hypotheses below C^1.** The preceding paragraph suggests the following hierarchy of assumptions. In the first two cases, the condition imposed at the outset is only membership in the relevant weak class; the zero-curvature equation is then a consequence of the convolution identity. The last option is not below C^1 in its Hölder version, and is best kept as a separate remark about a regime where one might also be able to proceed.

(1) *Convex surface of bounded extrinsic curvature.* Work in the Aleksandrov–Pogorelov category of convex hypersurfaces with bounded extrinsic curvature. In graph language, this means that the relevant Gauss–Kronecker curvature measure of the graph is defined. The preceding viscosity argument then forces this measure to vanish. This is the closest low-regularity version of the Hartman–Nirenberg/Pogorelov mechanism; it allows corners a priori and is therefore genuinely below C^1.

In this class, Pogorelov's developability theorem applies directly: a convex hypersurface of bounded extrinsic curvature whose curvature measure vanishes is developable. In the graph setting this means the following. For every point $(x, \psi(x))$ of the graph there is a nontrivial affine segment contained in the graph and passing through $(x, \psi(x))$. If the graph is complete and one uses the global form of the theorem, the local segments are the generators of a ruled hypersurface; away from the affine pieces these generators can be continued until they either meet the boundary or become complete affine lines. Since our graph is entire, there is no finite boundary in the base, so this is the natural route to the desired complete rulings.

(2) *Monge–Ampère function with generalized determinant.* Require that ψ belongs to a class of Monge–Ampère functions in which the Hessian determinant is defined weakly and continuously. The argument above should then give

$$\operatorname{Det} D^2\psi = 0$$

as a measure/current, rather than requiring this as an assumption. In dimension two, Jerrard's rigidity theorem for Monge–Ampère functions gives developability under such a weak formulation [8]. The point for us would be to use the analogous higher-dimensional determinant or Monge–Ampère system, if available, as a substitute for pointwise differentiability.

Even without quoting the full Pogorelov language, one gets the local ruled conclusion from the Aleksandrov maximum principle. Indeed, after subtracting a supporting affine function at x_0, suppose that the contact set of ψ with that support were reduced to x_0 inside a small ball. Then a small section

$$S_h := \{x : \psi(x) < h\}$$

would be a bounded convex set with $x_0 \in S_h$ and $\psi = h$ on ∂S_h. Aleksandrov's maximum principle bounds the height of such a section from above by a dimensional constant times the Monge–Ampère mass of S_h. Since $\operatorname{Det} D^2\psi = 0$, this mass is zero, forcing $h = 0$, a contradiction. Thus every contact set contains a second point, and the convexity of the contact set gives an affine segment in the graph through $(x_0, \psi(x_0))$.

This proves the local ruling statement in the weak Monge–Ampère class. To promote these local contact segments to complete affine lines in the full $d \geq 3$ problem, one still needs the corresponding global developability theorem in that weak class. Jerrard supplies this mechanism in dimension two; a higher-dimensional analogue would give the required replacement for Pogorelov in option (1).

Proposition 7 (Local rulings from zero Monge–Ampère measure)**.** *Let $\Omega \subset \mathbb{R}^d$ be open and convex, and let $u : \Omega \to \mathbb{R}$ be convex with Aleksandrov Monge–Ampère measure $M_u = 0$ in Ω. Let $x_0 \in \Omega$ and let $p \in \partial u(x_0)$. Then the contact set*

$$C(x_0, p) := \{x \in \Omega : \ u(x) = u(x_0) + p \cdot (x - x_0)\}$$

is not reduced to $\{x_0\}$ in any compactly contained convex neighborhood of x_0. Consequently the graph of u contains a nontrivial affine segment through $(x_0, u(x_0))$.

Proof. Subtract the supporting affine function and assume

$$u(x_0) = 0, \qquad u \geq 0, \qquad p = 0.$$

Suppose, toward a contradiction, that for some bounded convex $U \Subset \Omega$ containing x_0 the contact set $\{u = 0\} \cap U$ is exactly $\{x_0\}$. Then $u > 0$ on ∂U. Choose

$$0 < h < \min_{\partial U} u.$$

The section

$$S_h := \{x \in U : \ u(x) < h\}$$

is a nonempty bounded convex set with compact closure in U, and the convex function

$$w := u - h$$

satisfies $w < 0$ in S_h and $w = 0$ on ∂S_h. Aleksandrov's maximum principle gives, for every $x \in S_h$,

$$|w(x)|^d \leq C_d \operatorname{diam}(S_h)^{d-1} \operatorname{dist}(x, \partial S_h)\, M_u(S_h).$$

But $M_u(S_h) = 0$, hence $w \equiv 0$ in S_h, contradicting $w(x_0) = -h < 0$. Therefore every such neighborhood contains a second contact point $x_1 \neq x_0$. Since $C(x_0, p)$ is convex, the segment $[x_0, x_1]$ lies in $C(x_0, p)$, and the corresponding segment of the graph is affine. □

Remark 8 (Other rigidity thresholds)**.** *There may also be a Sobolev or fractional C^1 route. The role of this option is different from the two alternatives above: it does not give a canonical curvature measure below C^1, but it places the weak homogeneous Monge–Ampère equation in a known rigidity regime. In the two-dimensional weak Monge–Ampère literature, developability for the degenerate equation is rigid above certain thresholds, while convex-integration flexibility appears below them.*

For example, Jerrard's theorem is a rough rigidity result. Lewicka–Pakzad's convex-integration work exhibits the opposite, flexible side of the theory [9]*. The later De Lellis–Pakzad theorem proves developability for very weak homogeneous Monge–Ampère solutions in the little Hölder class $c^{1,2/3}$, hence for $C^{1,\alpha}$ with $\alpha > 2/3$* [4]*. In the Sobolev direction, a natural hypothesis to test is*

$$\psi \in W^{2,2}_{\mathrm{loc}}.$$

Then the equation $\det D^2\psi = 0$ *can be interpreted weakly, and the same section-law argument above should force that weak equation. If one then has a developability theorem in this Sobolev class, the conclusion is again that the graph is locally ruled by affine segments. A higher-dimensional version would need enough Sobolev control to make the nullity distribution of* $D^2\psi$ *integrable; this is precisely the step that fails in the flexible convex-integration regime. Thus this third route is plausible, but it should be treated as a rigidity-threshold alternative rather than as the primary low-regularity convex hypothesis.*

4.3. **Horizontal rulings and the final volume contradiction.** The nonnegativity normalization gives the last geometric restriction on a complete ruling. Suppose that the graph contains a complete affine line

$$\ell(s) = (x_0 + sv, \psi(x_0) + s\alpha), \qquad s \in \mathbb{R},$$

with $v \in \mathbb{R}^d$ and $\alpha \in \mathbb{R}$. Since ℓ lies in the graph, one has

$$\psi(x_0 + sv) = \psi(x_0) + s\alpha \qquad \text{for every } s \in \mathbb{R}.$$

If $\alpha \neq 0$, then the right-hand side is negative for one of the two limits $s \to +\infty$ or $s \to -\infty$, contradicting $\psi \geq 0$. Hence

$$\alpha = 0.$$

Thus every complete ruling is horizontal, i.e. parallel to the base $\mathbb{R}^d \times \{0\}$, and ψ is constant along the corresponding complete line in the base.

This observation closes the argument as soon as the developability theorem gives a cylindrical conclusion, namely a nontrivial linear subspace $L \subset \mathbb{R}^d$ such that

$$\psi(x + z) = \psi(x) \qquad \text{for every } x \in \mathbb{R}^d,\ z \in L.$$

Indeed, write $\mathbb{R}^d = L \oplus L^\perp$. For $z \in L$ and $\eta \in L^\perp$, the invariance gives

$$\Psi(2x, z + \eta) = \psi(x + \eta) + \psi(x - \eta) - 2\psi(x).$$

Consequently, for each $t > 0$,

$$\{z + \eta :\ z \in L,\ \eta \in L^\perp,\ \psi(x + \eta) + \psi(x - \eta) - 2\psi(x) < t\} \subset \{y : \Psi(2x, y) < t\}.$$

The transverse set in $L^\perp$ contains a neighborhood of 0 by continuity of ψ, while L has infinite Lebesgue measure. Hence

$$|\{y : \Psi(2x, y) < t\}| = +\infty \qquad \text{for every } t > 0,$$

contradicting the exact section law (2).

Thus the final higher-dimensional contradiction is reduced to the following purely geometric point: the complete horizontal rulings obtained from zero curvature should be parallel, or more generally should contain a common nontrivial translation subspace L along which ψ is invariant. This is exactly what the Hartman–Nirenberg cylindrical conclusion gives in the smooth complete case. In the low-regularity alternatives above, this is the global developability/cylinder statement that must be supplied. Mere existence of one complete horizontal line through each graph point gives only a zero-measure line inside each centered section; parallelism is what upgrades this to an infinite-volume product.

Proposition 9 (Cylindrical obstruction)**.** *Let $d \geq 3$, let $\psi : \mathbb{R}^d \to [0, \infty)$ be finite and convex, and assume* (2)*. Suppose that there is a nontrivial linear subspace $L \subset \mathbb{R}^d$ such that*

$$\psi(x + z) = \psi(x) \qquad \text{for every } x \in \mathbb{R}^d,\ z \in L. \tag{8}$$

Then (2) *is impossible.*

Proof. Fix $x \in \mathbb{R}^d$ and $t > 0$. Choose an orthogonal splitting $\mathbb{R}^d = L \oplus L^\perp$. For $z \in L$ and $\eta \in L^\perp$, (8) gives

$$\Psi(2x, z + \eta) = \psi(x + z + \eta) + \psi(x - z - \eta) - 2\psi(x) = \psi(x + \eta) + \psi(x - \eta) - 2\psi(x).$$

The function

$$\eta \mapsto \psi(x + \eta) + \psi(x - \eta) - 2\psi(x)$$

is continuous on $L^\perp$ and equals 0 at $\eta = 0$. Hence there is $\rho > 0$ such that it is $< t$ whenever $\eta \in L^\perp$ and $|\eta| < \rho$. Thus, for every $R > 0$,

$$\{z + \eta :\ z \in L,\ |z| < R,\ \eta \in L^\perp,\ |\eta| < \rho\} \subset \{y : \Psi(2x, y) < t\}.$$

By the product decomposition of Lebesgue measure along $L \oplus L^\perp$, the left-hand side has measure

$$\omega_{\dim L} R^{\dim L}\, \omega_{d-\dim L} \rho^{d-\dim L}.$$

Letting $R \to \infty$ shows that

$$|\{y : \Psi(2x, y) < t\}| = +\infty,$$

contradicting (2). □

Corollary 10 (Conclusion under the Pogorelov alternative)**.** *Let $d \geq 3$, let $\psi : \mathbb{R}^d \to [0, \infty)$ be finite and convex, and assume* (2) *and shrinking centered sections. Assume moreover that the graph of ψ belongs to the Aleksandrov–Pogorelov bounded extrinsic curvature class and that the corresponding global developability theorem gives the cylindrical conclusion* (8) *whenever the curvature measure vanishes. Then no such ψ exists.*

Proof. By the viscosity argument in the preceding subsection, the convolution identity forces the Aleksandrov Monge–Ampère measure to vanish. In the Aleksandrov–Pogorelov class this is equivalent, through the spherical-image parametrization of the graph, to vanishing extrinsic curvature measure on the graph. The assumed global Pogorelov developability theorem then gives a nontrivial common translation subspace L satisfying (8). Proposition 9 contradicts (2). □

Corollary 11 (Conclusion under the weak Monge–Ampère alternative)**.** *Let $d \geq 3$, let $\psi : \mathbb{R}^d \to [0, \infty)$ be finite and convex, and assume* (2) *and shrinking centered sections. Assume moreover that ψ belongs to a weak Monge–Ampère class in which:*

1. *the generalized determinant agrees with the Aleksandrov Monge–Ampère measure for convex functions;*
2. *zero generalized determinant implies the global cylindrical conclusion* (8)*.*

Then no such ψ exists.

Proof. The argument above gives

$$\operatorname{Det} D^2\psi = 0$$

as a measure/current. Proposition 7 gives the local ruled conclusion at every point. The second assumption in the present corollary is the global rigidity input that promotes these local contact segments to a common cylindrical translation subspace $L \neq \{0\}$. Proposition 9 then gives the contradiction. □

For the present convolution problem, the first option seems best aligned with convexity. The regularity condition should be stated as membership in a curvature-measure class for the graph, rather than as $\det D^2\psi = 0$ a.e.; the latter misses singular Monge–Ampère mass, exactly as in the example $|x_1| + \frac{1}{2}|x|^2$ discussed earlier. If the forced vanishing of this measure implies a global cylindrical ruling, the horizontal ruling argument above rules out (2) for $d \geq 3$. The minimum regularity condition I would try first is therefore:

ψ finite convex on $\mathbb{R}^d$,
the graph of ψ lies in the Aleksandrov–Pogorelov
bounded extrinsic curvature class.

The convolution identity should then force the corresponding curvature measure to be zero. This is milder than C^1, keeps the singular curvature mass visible, and is precisely tailored to the mechanism that should produce rulings.

Remark 12. *This is the point at which the two earlier drafts fit together. The transcript correctly identified that the implication*

$$\det D^2\psi = \textit{constant a.e.}$$

does not exclude singular Alexandrov–Monge–Ampère mass. The viscosity comparison above is stronger: upper supports see the missing lower bound for the determinant, and lower supports see the upper bound. Thus no separate uniqueness theorem for quadratic blowups is needed once the exact model section law is available.

Remark 13. *For comparison, a positive definite quadratic graph in dimension d has centered convolution proportional to $\tau^{d/2-1}$. Thus literal constancy on the normalized support is compatible with a nondegenerate quadratic model only when $d = 2$.*

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